

Stress-Testing Product-Gated Clustering and Bounded Priority Ranking for Flood-Rescue Reports: A Synthetic Study

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Abstract. Operational flood streams contain duplicate, incomplete, and adversarial reports, while clustering and priority scores are often evaluated without measuring their scheduling consequences. We audit a fail-closed batch pipeline in which observable reports form a sparse graph, unresolved reports go to review, duplicate families feed a bounded score, and evaluator-only truth is joined after predicted-cluster scheduling. The clustering benchmark contains 80 synthetic runs from generator 3.0.0: using Louvain community detection, an additive-affinity graph has mean ARI 0.9191, while the spatially gated product-affinity graph has 0.9165 with the Leiden community detector and 0.9072 with Louvain. Product minus independently tuned additive is -0.0118 (bootstrap 95% CI $[-0.0226, -0.0024]$), but its Holm-adjusted test is not significant ($p = .2560$). A matched-density comparison reverses the point estimate but is also inconclusive. Under fixed configurations, location and timestamp noise reduce clustering accuracy, while fivefold exact transport copies reduce all four graph variants to mean ARI near .30. In a separate candidate-4.1 bundle, the revised score has NDCG@5 0.6669 versus 0.6763 for a random comparator and does not show a general alignment advantage; its family-wise score is invariant to exact copies when evidence grouping is fixed, but coordinated high-confidence campaigns remain a failure mode. Dispatch results similarly show no general benefit over the legacy score and a clear cost relative to nearest-first. These are synthetic, within-generator diagnostics: boundedness and clustering accuracy do not establish source authenticity, policy validity, field benefit, or deployment readiness.

Keywords: Flood rescue · Graph clustering · Synthetic stress test · Priority ranking · Negative results.

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1 Introduction and Related Work

During emergencies, crisis streams mix repeated reports of one event with nearby but distinct events, missing fields, and nonincident noise. Social-sensor and emergency-message research addresses volume, verification, and event extraction [11]. CrisisMMD and FloodNet capture useful disaster modalities, although FloodNet is an aerial post-flood scene-understanding dataset rather than a multimodal report stream [1,10]. These resources do not jointly provide physical-incident linkage, rescue priority, and field dispatch outcomes.

Social sensing and event extraction. Social-media users can act as real-time sensors [11], but the usual endpoint is event detection or classification rather than the consequences of sending a field resource to a predicted incident.

Consolidation and clustering. Graph community detection with Louvain or Leiden [3,15], and density methods such as DBSCAN and HDBSCAN [5] offer plausible consolidation mechanisms. Multiplicative spatial-range terms occur in bilateral filtering [14]; we adapt that structure to a product graph, while testing the exact gate and its bounds here. Throughout the paper, *Product Louvain* denotes Louvain applied to a graph whose temporal and contextual affinity is multiplied by a spatial gate; *Product Leiden* keeps that graph and changes only the community detector. *Additive Louvain* instead sums the spatial, temporal, and contextual terms. The *matched-density Additive* diagnostic adjusts its sparsification settings without labels to approximate the Product graph’s retained-edge fraction and mean degree.

Humanitarian logistics and dispatch. Relief-routing models expose competing equity, efficacy, and efficiency objectives [6]. We use this perspective to motivate a bounded priority ranking, but boundedness is not an expert-endorsed rescue policy. A key gap is propagation of false splits, merges, and noise into a scheduler that cannot consult incident truth.

We address these gaps with a fail-closed batch pipeline: reports with observable location and time enter a sparse graph, incomplete reports go to manual review, duplicate families feed a bounded ranking, and evaluator-only truth joins after predicted-cluster scheduling. We ask: (RQ1) does independently selected product composition improve clustering over additive composition? (RQ2) does the duplicate-aware score align with an independently generated benefit under report-level stress tests? (RQ3) how do predicted split, merge, and noise errors propagate into dispatch? The questions use two explicitly separated artifact bundles.

The contributions are (i) an observable-only pipeline with explicit review and evaluator boundaries; (ii) auditable threshold statements and a proposed duplicate-aware score; and (iii) a paired diagnostic reporting clustering, stress behavior, ranking, dispatch, and adverse results. The practical value is diagnostic: the spatial gate supplies a checkable edge-localization condition, family-wise

aggregation supplies an exact-copy invariance test under fixed evidence grouping, and the downstream evaluation exposes when strong clustering scores fail to improve dispatch. These safeguards can serve as implementation checks, but they do not establish superiority or field utility. The evidence remains synthetic and within-generator, so the artifact is not a validated rescue policy.

2 Observable Pipeline and Methods

Figure 1 fixes the information boundary. Receipt-visible fields flow through graph construction, ranking, and a batch scheduler. Missing location or event time routes a report to review. Incident identity, benefit, deadline, and harm are stored separately and join only after a schedule exists.

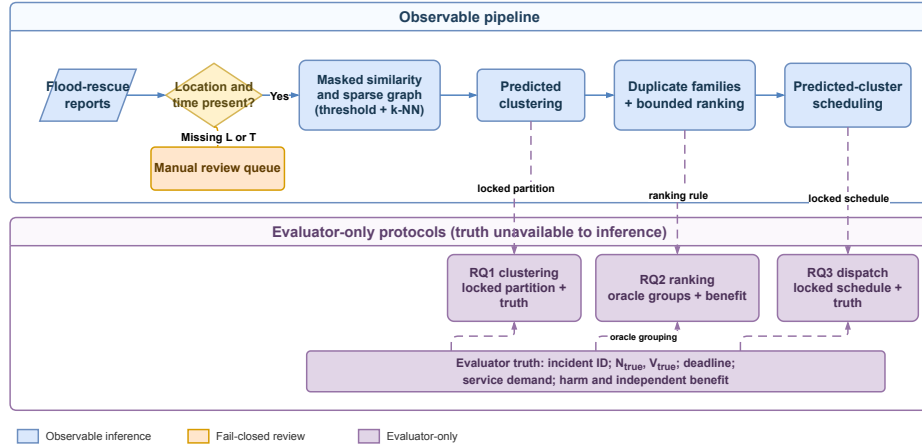


Fig. 1. Observable pipeline (solid), manual-review path (orange), and evaluator-only truth join (dashed). The scheduler receives predicted clusters, never incident truth.

Algorithms 1 and 2 restate admission, graph construction, duplicate-aware scoring, and scheduling; neither receives the evaluator-only truth join.

2.1 Report Attributes and Missingness

Each report is represented by the compact tuple

$$r_i = (L_i, T_i, F_i, E_i, N_i, V_i, Q_i),$$

where L_i is a GPS location and T_i is the observable report timestamp (“created_at”). Flood evidence F_i and urgency evidence E_i are in $[0, 1]$; $N_i \in \mathbb{Z}_{\geq 0}$ is the reported trapped-person count; $V_i \geq 0$ is a vulnerability-evidence value; and

$Q_i \in [0, 1]$ is a provenance/confidence heuristic. Q_i is not a calibrated probability of truth. The implementation also carries observable support fields such as image presence, source type, province, and missing-field flags; these support Q_i and duplicate fingerprints but are not additional latent labels.

The score Q_i is computed from observable image presence and the number of nearby corroborating observable payloads (counted by unique payload identity) using the proposed heuristic

$$Q_i = \text{sigmoid}(b_0 + b_1 \mathbf{1}\{\text{image}_i\} + b_2 \log(1 + n_i^{\text{corrob}})), \quad (1)$$

$$(b_0, b_1, b_2) = (-.2, 1.4, .9).$$

Here $\text{sigmoid}(x) = (1 + e^{-x})^{-1}$, $\mathbf{1}\{\text{image}_i\}$ is one exactly when report i contains an image, and $n_i^{\text{corrob}} \in \mathbb{Z}_{\geq 0}$ counts distinct nearby observable payloads. The dimensionless b_0, b_1, b_2 are the fixed intercept, image, and corroboration coefficients; consequently $Q_i \in (0, 1)$. This is a pipeline rule, not a calibrated truth model. No independent receipt timestamp or source family is assumed.

Following the general missing-variable kernel problem [12], we use the following fail-closed construction. The admission indicator $a_i = \mathbf{1}\{L_i, T_i \text{ are observed}\}$. Thus $a_i a_j = 0$ prevents an edge involving a report whose geographic or temporal separation is undefined; such a report follows the manual-review path. Missing F_i or E_i is not zero-imputed: let $I_F(i, j), I_E(i, j) \in \{0, 1\}$ indicate that both reports contain the corresponding observation and let $m_{ij} = I_F(i, j) + I_E(i, j)$; then

$$C_{ij} = \frac{m_{ij}}{2} \exp\left[-I_F(i, j) \frac{|F_i - F_j|}{\tau_F} - I_E(i, j) \frac{|E_i - E_j|}{\tau_E}\right], \quad C_{ij} = 0 \text{ if } m_{ij} = 0. \quad (2)$$

$\tau_F, \tau_E > 0$ are dimensionless flood- and urgency-difference scales, so $C_{ij} \in [0, 1]$. When $m_{ij} = 2$, both contextual fields are used; when $m_{ij} = 1$, the maximum contextual similarity is discounted to $1/2$; and when $m_{ij} = 0$, no contextual agreement is inferred. In the current 80-run artifact, every report has location, event time, flood, and urgency. Hence $a_i = 1$ and $m_{ij} = 2$ throughout the evaluated data, so Eq. 2 uses its full two-field exponential branch. The missingness extension is specified and unit-tested, but missingness robustness is not claimed as an empirical result of this benchmark.

2.2 Similarity Graph and Derived Bounds

For Haversine distance d_{ij} and time difference Δt_{ij} , define the Gaussian affinity [8] and exponential temporal decay [18]:

$$G_{ij} = \exp[-d_{ij}^2/(2\sigma^2)], \quad T_{ij} = \exp[-\Delta t_{ij}/\tau_t],$$

$$w_{ij}^\times = a_i a_j G_{ij} (\beta T_{ij} + \gamma C_{ij}), \quad w_{ij}^+ = a_i a_j (\alpha G_{ij} + \beta T_{ij} + \gamma C_{ij}). \quad (3)$$

d_{ij} is measured in metres and Δt_{ij} in minutes; $\sigma > 0$ and $\tau_t > 0$ are the corresponding spatial and temporal bandwidths. The dimensionless $G_{ij}, T_{ij}, C_{ij} \in [0, 1]$ enter the product and additive weights $w_{ij}^\times$ and w_{ij}^+ with nonnegative α, β, γ .

Superscripts $\times$ and $+$ identify composition, not multiplication or addition over report indices. The exponential form is a precedent, not a claim that the complete missingness-aware expression is copied from prior work [2]. The product is a spatial gate motivated by spatial-range products such as bilateral filtering [14]; both compositions and all weights here are defined by this paper. The implementation forms the eligible pair matrix, computes an empirical nonzero-weight quantile, removes weights at or below that threshold, retains the endpoint-wise top- k among the remaining directed neighbours, and symmetrizes their union before community detection. Thus an undirected node may have degree above k when another node selects it. Product and additive candidates are selected independently; the estimand is therefore a pipeline comparison, not an operator-only causal effect. We treat this independently calibrated contrast as the primary estimand for RQ1 because it asks whether each tuned pipeline provides a net empirical benefit. Section 4 also reports a matched-density contrast as a secondary diagnostic. It reduces differences in retained-edge fraction and mean degree, but different edge weights and memberships remain, so it does not identify a causal effect of the composition operator.

Theorem 1 (Product edge localization). *Let $B = \beta + \gamma > 0$. If a product edge with threshold $0 < \theta < B$ is retained, then $d_{ij} < \sigma\sqrt{2\log(B/\theta)}$. For $\theta \geq B$ the strict retained set is empty; for $\theta \leq 0$ the threshold alone gives no finite cutoff.*

Proof. Since $a_i a_j \leq 1$ and $T_{ij}, C_{ij} \leq 1$, $w_{ij}^\times \leq B \exp[-d_{ij}^2/(2\sigma^2)]$. Combining $\theta < w_{ij}^\times$ with positivity and taking logarithms gives the bound; the endpoint cases follow from the same inequality.

This is an edge statement. A component-diameter claim additionally needs a bounded observed hop diameter; long transitive chains remain possible.

Corollary 1 (Conditional component bound). *In the finite product region, a connected component with at least two nodes, unweighted observed hop diameter $1 \leq h < \infty$, and geographic diameter D satisfies $D < hr_\theta^\times$, where $r_\theta^\times = \sigma\sqrt{2\log(B/\theta)}$. A singleton has $h = D = 0$ and is handled separately.*

For a non-singleton, the result follows from the triangle inequality along a path of at most h retained edges. It is a conditional post-hoc component bound, not an ex-ante compactness guarantee: the pipeline does not control h .

For the additive form, assume $\alpha > 0$. If $B < \theta < B + \alpha$, every retained edge satisfies

$$d_{ij} < r_\theta^+ = \sigma\sqrt{2\log\left(\frac{\alpha}{\theta - B}\right)}.$$

If $\theta \geq B + \alpha$, the strict retained set is empty; if $\theta \leq B$, no non-trivial threshold-induced geographic cutoff exists in general. Indeed, retention gives $\theta < \alpha G_{ij} + \beta T_{ij} + \gamma C_{ij} \leq \alpha G_{ij} + B$. Thus product composition has a finite geographic edge bound throughout its nonempty positive-threshold domain, whereas additive composition has such a bound only above the maximum non-geographic contribution. Neither statement alone implies better empirical clustering.

Algorithm 1 Fail-closed observable report pipeline (Fig. 1)

Require: Reports $R = \{r_i\}$; kernel scales $\sigma, \tau_t, \tau_F, \tau_E$;
weights β, γ (and α for the additive composition); quantile q ; neighbour cap
 k ; Louvain/Leiden resolution ρ ; composition $m \in \{\text{product, additive}\}$
Ensure: Dispatch schedule S ; the algorithm below never reads incident
identity, benefit, deadline, or harm labels

- 1: $R_{\text{obs}} \leftarrow \emptyset$; $R_{\text{review}} \leftarrow \emptyset$
- 2: **for** each report $r_i \in R$ **do**
- 3: $a_i \leftarrow \mathbf{1}\{L_i, T_i \text{ observed}\}$ ▷ Section 2.1
- 4: **if** $a_i = 0$ **then**
- 5: $R_{\text{review}} \leftarrow R_{\text{review}} \cup \{r_i\}$ ▷ orange manual-review path
- 6: **else**
- 7: $R_{\text{obs}} \leftarrow R_{\text{obs}} \cup \{r_i\}$
- 8: **end if**
- 9: **end for**
- 10: Compute n_i^{corrob} from unique payloads in R_{obs} and set every Q_i by Eq. 1
- 11: $E_0 \leftarrow \text{BUILDSIMILARITYEDGES}(R_{\text{obs}}, m)$
using the fixed scales and weights above
BuildSimilarityEdges computes G, T, C on the full eligible pair matrix
- 12: $\theta \leftarrow q$ -quantile of the nonzero weights in E_0
- 13: $E_\theta \leftarrow \{(i, j, w_{ij}) \in E_0 : w_{ij} > \theta\}$
- 14: $E^\rightarrow \leftarrow \{(i, j, w_{ij}) \in E_\theta : j \text{ is among } i\text{'s top-}k \text{ neighbours in } E_\theta\}$
- 15: $E \leftarrow \{(i, j) : (i, j) \in E^\rightarrow \text{ or } (j, i) \in E^\rightarrow\}$
- 16: $\mathcal{G} \leftarrow (R_{\text{obs}}, E)$
- 17: $\mathcal{C} \leftarrow \text{Louvain/Leiden}(\mathcal{G}, \rho)$ ▷ predicted clusters, never incident truth
- 18: **for** each predicted cluster $C_k \in \mathcal{C}$ **do**
- 19: $P_k \leftarrow \text{PRIORITYSCORE}(C_k)$ ▷ Algorithm 2
- 20: **end for**
- 21: $S \leftarrow \text{Scheduler}(\{(C_k, P_k) : C_k \in \mathcal{C}\})$
scheduler sees predicted clusters and P_k only
- 22: **return** S

After S is fixed, the offline evaluator may join S and R_{review} with incident identity, benefit, deadline, and harm labels to compute ARI, NDCG@5, and dispatch outcomes. This dashed path is not available to the inference algorithm. In the current artifacts, the core schema requires location and event time, so the manual-review branch is a fail-closed design boundary rather than an empirically exercised missingness condition.

2.3 Duplicate-Aware Bounded Ranking

Exact duplicate payloads share a canonical fingerprint. Near duplicates are joined by deterministic connected components under the observable envelope; this transitive rule is not complete linkage. For a predicted cluster C_k , let $\mathcal{F}_k$ denote the resulting exact/near-duplicate evidence families inside that cluster. The revised estimator uses one confidence-gated priority per cluster and family

maxima so that overlapping reports are not treated as disjoint people. Write $\tilde{N}_i = \min(N_i, N_{\text{ref}})$ and $\tilde{V}_i = \min(V_i, V_{\text{cap}})$ for the nonnegative clipped claims:

$$\begin{aligned}\bar{E}_k &= \frac{\sum_{g \in \mathcal{F}_k} \max_{i \in g} Q_i E_i}{\max(1, \sum_{g \in \mathcal{F}_k} \max_{i \in g} Q_i)}, & \bar{F}_k &= \max_{g \in \mathcal{F}_k} \max_{i \in g} Q_i F_i, \\ \bar{N}_k &= \min\left(1, \frac{\log(1 + \max_{g \in \mathcal{F}_k} \max_{i \in g} Q_i \tilde{N}_i)}{\log(1 + N_{\text{ref}})}\right), & \bar{V}_k &= \max_{g \in \mathcal{F}_k} \max_{i \in g} Q_i \tilde{V}_i, \\ P_k &= [1 + (\mu - 1) \tanh(\bar{V}_k/s)] (\omega_E \bar{E}_k + \omega_F \bar{F}_k + \omega_N \bar{N}_k).\end{aligned}\tag{4}$$

g indexes an evidence family and $i \in g$ a report in that family. $\bar{E}_k$ is a confidence-gated family aggregation, while $\bar{F}_k$, $\bar{N}_k$, and $\bar{V}_k$ are respectively maximum flood, normalized clipped population, and clipped vulnerability evidence. Here $N_{\text{ref}} > 0$ is the population reference, $V_{\text{cap}} \geq 0$ the vulnerability cap, $s > 0$ the saturation scale, $1 \leq \mu \leq 2$ the maximum amplification, and $\omega_E, \omega_F, \omega_N \geq 0$ sum to one. Therefore $\bar{E}_k, \bar{F}_k, \bar{N}_k \in [0, 1]$, $\bar{V}_k \in [0, V_{\text{cap}}]$, and $0 \leq P_k \leq \mu$. The candidate source snapshot associated with the RQ2/RQ3 artifacts specifies $\omega = (.34, .33, .33)$, $\mu = 2$, $s = 10$, $N_{\text{ref}} = 500$, and $V_{\text{cap}} = 50$. Maxima limit duplicate multiplicity, the logarithm compresses demand, clipping bounds individual evidence, and the hyperbolic tangent saturates vulnerability amplification. Reliability-aware truth discovery motivates Q_i but does not derive these choices [16]; they still require sensitivity and expert validation [13,6]. Comparators test raw multiplicity (legacy), single factors (urgency/population), an unbounded linear score, chance ordering, and travel time (nearest-first); none is an endorsed rescue policy.

Algorithm 2 PRIORITYSCORE: duplicate-aware bounded priority (Eq. 4)

```

1: function PRIORITYSCORE(cluster  $C_k$ )
2:    $\mathcal{F}_k \leftarrow \text{ExactFingerprintGroups}(C_k)$  and  $\text{NearDuplicateComponents}(C_k)$ 
3:   Compute  $\bar{E}_k, \bar{F}_k, \bar{N}_k, \bar{V}_k$  by Eq. 4
4:   Compute  $P_k$  by Eq. 4
5:   return  $P_k$ 
6: end function

```

3 Experimental Design

Data contract and splits. The study uses two explicitly separated artifact suites. RQ1 uses the tracked generator-3.0 bundle, a semi-synthetic dataset with geographic anchors from the Copernicus EMSR848 Central Viet Nam activation [4]; Copernicus EMS, OpenStreetMap extracts, and the 2025 WorldPop constrained population surface [17] provide geographic or historical anchors, while incidents, reports, duplicates, coordinated attacks, vulnerability, and labels are synthetic. It contains 80 runs with 16 latent incidents and 26,288 reports (17,901 genuine,

3,587 duplicate, and 4,800 coordinated nonincident reports), with 288–370 reports per run. Runs 1–20 are development, 21–40 calibration, and 41–80 the test split held out from calibration. These are within-generator tests, not mechanism-shift OOD transfer.

RQ2 and RQ3 use a separately generated Candidate-4.1 bundle. This bundle is a fully synthetic regional scenario defined by the generator and is not claimed to be EMSR848-anchored. It contains 80 datasets, 16 incidents per dataset, 30,229 reports in total, and 339–424 reports per dataset; the development, calibration, and test seeds are 1000–1019, 2000–2019, and 3000–3039. Across its splits there are 1,554 fake reports and 2,560 labeled duplicates (1,280 exact and 1,280 near). Each base dataset contains one five-report coordinated high-confidence campaign and one report for each of four low-confidence attack types; RQ2 stress scenarios perturb these inputs. For both suites, inference reads only observable `algorithm_input.json`; ground truth is opened only after predictions are fixed.

Fixed clustering stresses. The 40 held-out generator-3.0 test runs are also evaluated under seven fixed conditions: the unmodified control; independent Gaussian location noise with 100 m or 300 m standard deviation on each local axis; independent Gaussian timestamp noise with 15 min or 60 min standard deviation; and exact transport copies at total multiplicity two or five, where only the observable report identifier changes. Five configurations selected before this stress evaluation are retained without tuning: Product Louvain, Additive Louvain, matched-density Additive, Product Leiden, and geo-time DBSCAN. Each method receives the same deterministic perturbation for a run and condition, yielding $40 \times 7 \times 5 = 1,400$ fits. These controlled levels characterize the tested pipeline and are not estimates of field measurement error or duplicate prevalence.

Selection and comparators. The current notebook evaluates complete small grids on the 20 calibration runs and selects mean ARI, breaking ties by standard deviation and canonical configuration JSON. Product and additive use fixed $\tau_F = .25$, $\tau_E = .35$, and $\beta = \gamma = .5$. The Product grid varies $\sigma \in \{700, 1200\}$ m, $\tau_t \in \{30, 60\}$ min, $q \in \{.90, .95\}$, and $k \in \{8, 16\}$; the Additive grid varies $\alpha \in \{1/3, 1\}$, the same σ, q, k sets, and fixes $\tau_t = 60$ min. Both fix resolution at 1.2. Selected product Louvain uses $\sigma = 700$ m, $\tau_t = 60$ min, quantile .90, $k = 8$, and resolution 1.2. Selected additive Louvain uses $\alpha = 1$, $\sigma = 700$ m, $\tau_t = 60$ min, quantile .95, $k = 8$, and resolution 1.2. Because selected quantiles differ, this is an independently calibrated pipeline comparison, not a matched-density operator comparison. Comparators are convex similarity, Product Leiden, geo-time DBSCAN, DBSCAN/HDBSCAN on scaled observable features, spatial agglomeration, and coordinate K-means. Geo-time DBSCAN lacks ST-DBSCAN’s non-spatial criterion. Together these variants separate composition and community detection, reduce the Product/Additive density imbalance, and supply density-, hierarchy-, and geometry-based controls.

Table 1. Artifact construction. Geographic anchors do not make simulated reports or outcomes into field observations.

Layer	Generator 3.0 (RQ1)	Candidate 4.1 (RQ2/RQ3)
Geographic context	EMSR848 context, OSM-derived anchors, and WorldPop cells	Synthetic regional geometry; no EMSR848 claim
Reports/linkage	Synthetic incidents, reports, duplicates, fake campaigns, and labels	Synthetic incidents, reports, duplicates, and attack cases
Operational truth	Need and vulnerability simulated; population is not trapped-person truth	Benefit, deadlines, service, access, and harm simulated
Representativeness	Within-generator geographic plausibility; no source snapshot or row lineage	Controlled stress cases; no field sample or transfer

Endpoints and inference. ARI on incident-linked reports [7] is the primary endpoint; NMI, pairwise F1, geographic diameter, runtime, false destinations, fake absorption, noise rejection, retained-edge fraction, and mean degree are descriptive or operational diagnostics. The benchmark includes a matched-density product-additive comparison. Seed-paired ARI differences use 10,000 bootstrap resamples, Wilcoxon signed-rank tests, and Holm correction; RQ2 and RQ3 paired comparisons use 5,000 bootstrap resamples with the locked analysis seed. The reported p-values use explicitly exported analysis families. RQ1 applies one Holm family to all nine Product-Louvain-versus-comparator ARI contrasts; the three product/additive/Leiden contrasts are the primary comparisons and the remaining baselines are secondary diagnostics. RQ2 uses five comparator-specific three-endpoint families for alignment (NDCG@5, Spearman, and top-five overlap) and one five-endpoint family per robustness scenario (priority drift, mean-rank drift, top-k churn, N-error, and V-error); false priority lift is descriptive. For RQ3, each fixed candidate/comparator/context block applies Holm across all 14 dispatch endpoints. The primary RQ3 revision aggregates the three resource scenarios within each seed, so each paired test uses 40 seed pairs; the original 120 seed-by-scenario analysis is retained as a diagnostic artifact. RQ2 alignment uses oracle incident groups only to isolate ranking, whereas RQ3 evaluates predicted clusters. The priority stress tests and dispatch evaluation use independent benefit, deadline, service, access-delay, and harm outputs; no clustering metric is substituted for those outcomes. For the fixed RQ1 stresses, ARI and pairwise F1 on the original incident-linked reports keep the evaluation unit comparable across copy multiplicities. Metrics over all perturbed payloads and contamination, false-destination, and graph-density diagnostics remain descriptive. Each stress condition is paired with its method’s control by run; 5,000 bootstrap resamples estimate the mean within-method change and its unadjusted 95% CI. These intervals describe stress-versus-control changes and are not tests of between-method superiority.

4 Results

Across the 40 held-out runs, neither graph composition has a statistically robust advantage. The duplicate-aware score removes exact-copy inflation under fixed grouping but fails under a coordinated high-confidence campaign. Its dispatch

policy shows no general benefit over the legacy score and is worse than nearest-first on harm and deadlines. The details follow.

Table 2 reports the held-out clustering summary.

Additive Louvain has the highest mean ARI. Product minus independently tuned additive is -0.01184 with CI $[-0.02257, -0.00240]$, with 9 wins, 19 ties, and 12 losses; raw $p = .0987$ and Holm $p = .2560$. Product minus matched-density additive is $+0.00989$ with CI $[-0.00100, 0.02082]$ and Holm $p = .2560$. Product Leiden versus Product Louvain has Holm $p = .1120$. Thus neither composition advantage is established. Product and matched-density Additive have nearly equal retained-edge fractions (.03011 and .03021) and mean degrees (9.90 and 9.94), so the matched-density contrast is a useful diagnostic but not a causal operator comparison.

The product pipeline has zero fake absorption and zero fake rejection but a 16.49% false-destination rate; fake reports avoid genuine destinations while still forming fake-only destinations. This is an adverse operational result, not evidence of misinformation robustness.

Figure 2 reports the fixed clustering stresses as paired changes from each method’s control. All tested location and timestamp noise levels reduce mean ARI, and no method has the highest mean ARI in every condition.

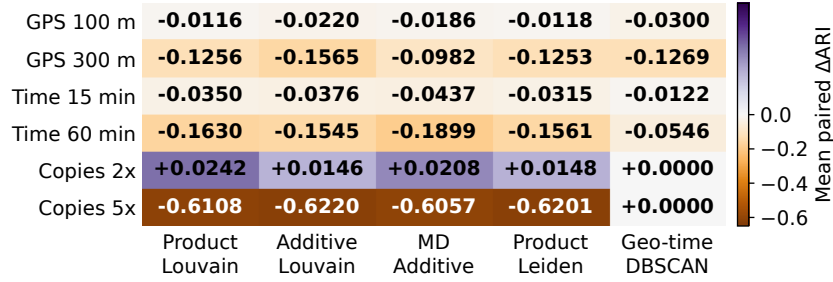
Twofold copies slightly increase ARI for all four graph variants, whereas fivefold copies reduce their means to approximately .30; geo-time DBSCAN remains at its lower control mean. For Product Louvain, the fivefold change is $-.6108$ with an unadjusted paired-bootstrap 95% CI $[-.6318, -.5881]$. Thresholding and endpoint-wise top- k sparsification are plausible mechanisms, but this protocol does not isolate a cause. For original-report ARI, an independent rerun under the same locked protocol changed only the aggregate twofold-copy graph values slightly; the main GPS, time-noise, fivefold-copy, and cross-condition conclusions were unchanged. The two runs are not pooled. Thus score-level invariance does not imply clustering invariance or end-to-end robustness.

For RQ2, revised NDCG@5 is .6669 (CI $[\.6306, .7018]$), Spearman is $-.0304$ (CI $[-.1165, .0549]$), and top-five overlap is .285 (CI $[\.2349, .3400]$). The revised score is above legacy on NDCG@5 by .0121, but the Holm-adjusted contrast is not significant; it is below population-only (.6709) and random (.6763). Exact duplication at $2\times$, $5\times$, and $10\times$ produces zero revised priority drift. A coordinated high-confidence campaign produces drift .1688 and false-priority lift .8828; revised drift is worse than legacy by $-.1379$ (Holm $p \approx 9.1 \times 10^{-12}$). Boundedness therefore does not guarantee ranking reliability.

Table 3 expands the robustness audit beyond the two headline cases. The revised score is invariant to exact-copy multiplicity and usually changes less under near duplication, missingness, contradictory evidence, and low-confidence flood inflation. Its bounded form is nevertheless more sensitive than the legacy score to low-confidence urgency, population, or vulnerability inflation and to coordinated high-confidence reports. These are controlled synthetic perturbations, not estimates of attack prevalence or field error.

Table 2. RQ1 benchmark-test summary over 40 runs. Intervals are bootstrap 95% confidence intervals; diameter is descriptive.

Method	ARI [95% CI]	Pair F1	Mean diam. (km)
Additive Louvain	.9191 [.8995,.9381]	.9247	.3477
Product Leiden	.9165 [.8979,.9351]	.9223	.3547
Convex Louvain	.9135 [.8940,.9320]	.9195	.3547
Product Louvain	.9072 [.8864,.9271]	.9138	.3585
Matched-density Additive	.8973 [.8763,.9175]	.9046	.3743
Coordinate K-means	.8176 [.7976,.8384]	.8292	.2977
Spatial agglomerative	.6235 [.5964,.6494]	.6506	4.1648
HDBSCAN	.5205 [.4565,.5811]	.5787	1.2527
Geo-time DBSCAN	.4978 [.4589,.5366]	.5450	.6244
DBSCAN, all features	.4741 [.4273,.5201]	.5488	.9340

**Fig. 2.** Mean paired change in RQ1 ARI relative to each method’s control over 40 held-out runs. Cells use original incident-linked reports; negative values indicate degradation; MD Additive denotes matched-density Additive. Intervals are not shown and no between-method test is implied.

For RQ3, after averaging the three resource scenarios within each seed, revised versus legacy on the Product partition has harm effect -14.34 (Holm $p = .563$) and deadline effect $+0.0188$ (Holm $p = .247$), so no general improvement is established. Revised is significantly worse than nearest-first on both harm (Holm $p = 4.1 \times 10^{-5}$) and deadline (Holm $p = 1.3 \times 10^{-6}$). Product is also worse than matched-density Additive under the revised policy on harm (Holm $p = .000168$), and worse than oracle on the corresponding dispatch outcomes. Predicted split, merge, and fake-only destinations therefore propagate into operational consequences.

Resource-specific Product-partition means (revised, legacy, nearest-first) for harm and deadline-miss rate were: lean Hue, 691.37, 653.79, 585.55 and .9188, .9203, .8203; nominal dual depot, 315.01, 303.16, 230.00 and .8453, .8594, .6313; and regional surge, 113.53, 119.93, 94.71 and .6672, .7078, .4484. Revised priority lowers mean deadline misses relative to legacy in the nominal and surge scenarios, and lowers harm only in surge; nearest-first is lower on both outcomes in every scenario. These means are descriptive. Primary inference still averages the three scenarios within each seed, giving 40 seed-level pairs rather than treating the 120 seed-scenario rows as independent.

Table 3. RQ2 robustness means over the held-out Candidate-4.1 test split. n counts target-perturbation rows; ΔP is absolute normalized priority drift, Δr is normalized mean-rank drift, and churn is top-five churn. R/L denote revised/legacy; lower is better.

Scenario	n	ΔP_R	ΔP_L	Δr_R	Δr_L	Churn _R	Churn _L
Exact duplicate 2×	40	.0000	.0026	.0000	.0015	.0000	.0000
Exact duplicate 5×	40	.0000	.0086	.0000	.0046	.0000	.0100
Exact duplicate 10×	40	.0000	.0148	.0000	.0075	.0000	.0150
Near duplicate	320	.0016	.0055	.0004	.0017	.0006	.0019
Contradictory F/E	40	.0484	.0611	.0171	.0290	.0450	.1100
Missingness/zero-imputation	628	.0063	.0075	.0019	.0025	.0029	.0064
Low-confidence E inflation	40	.0034	.0012	.0008	.0006	.0050	.0000
Low-confidence F inflation	40	.0026	.0072	.0000	.0040	.0000	.0150
Low-confidence N inflation	40	.0696	.0140	.0240	.0071	.0600	.0150
Low-confidence V inflation	40	.0758	.0074	.0263	.0040	.0500	.0150
Coordinated high-confidence	40	.1688	.0309	.0175	.0006	.0350	.0000

4.1 Sensitivity of Heuristic Constants

To assess robustness to the fixed constants in Q_i (Section 2.1) and P (Eq. 4), we define a local one-at-a-time sensitivity analysis on the Candidate-4.1 ranking task. Each valid constant is perturbed around its reference value while all other constants are held fixed, and NDCG@5 is recomputed on the same 40 test seeds. The ω variants perturb one component at a time and renormalize the vector; μ is restricted to its policy-valid interval $[1, 2]$. The locked rerun evaluates 21 configurations (one shared baseline and two or three tested levels for each constant) on the same 40 held-out seeds, yielding 840 seed-level observations. A constant is considered stable if the mean metric varies by less than 0.01 across its valid tested levels. The sensitivity summary CSV provides the ranges below; no value is inferred from the fixed-configuration comparisons.

Table 4 reports these ranges.

5 Discussion and Threats to Validity

The results separate mathematical safeguards from empirical utility. Product gating supplies a finite edge bound, but neither the independently tuned nor matched-density comparison establishes a preferred composition. Location, timestamp, and exact-copy stresses expose further clustering failures without identifying thresholding or top- k selection as their cause. Likewise, score invariance requires a fixed evidence-family partition and fixed inputs; it does not constrain an upstream graph changed by transport copies. A bounded score can still shift under a coordinated campaign, and cannot substitute for a validated operational objective when clustering errors and fake-only destinations reach scheduling.

The local sensitivity analysis is stable over the tested intervals: every constant has a mean-NDCG@5 span below 0.01 across the 40 held-out seeds. The largest

Table 4. Candidate-4.1 one-at-a-time sensitivity of mean NDCG@5 to fixed heuristic constants. Each range is the minimum–maximum mean over the tested levels, including the reference level, on 40 test seeds. All spans are below 0.01 under the locked rerun.

Constant	Reference/default value	Tested levels	Metric range
b_0 (Q_i intercept)	−0.20	[−0.24, −0.16]	[.6653, .6672]
b_1 (image weight)	1.40	[1.12, 1.68]	[.6669, .6687]
b_2 (corroboration weight)	0.90	[0.72, 1.08]	[.6667, .6679]
ω_E	.34	component $\times \{.8, 1, 1.2\}$, renormalized	[.6663, .6714]
ω_F	.33	component $\times \{.8, 1, 1.2\}$, renormalized	[.6625, .6679]
ω_N	.33	component $\times \{.8, 1, 1.2\}$, renormalized	[.6663, .6697]
μ (vulnerability amplification)	2	$\{1.6, 1.8, 2.0\}$	[.6644, .6683]
s (vulnerability saturation)	10	[8, 12]	[.6665, .6713]
N_{ref}	500	[400, 600]	[.6669, .6669]
V_{cap}	50	[40, 60]	[.6669, .6669]

span is for ω_F (.00546), while changing V_{cap} from 40 to 60 leaves the mean unchanged at the reported precision. These results support numerical robustness within the tested neighborhood only; they do not validate the constants outside those levels or replace expert validation.

All reports, incidents, relevance signals, and dispatch outcomes are synthetic. RQ1 varies seeds within generator 3.0.0, while RQ2/RQ3 use a separate candidate-4.1 bundle; neither tests mechanism-shift OOD transfer or field performance. RQ1 does not provide an empirical missingness test because all evaluated reports contain the four core attributes. No expert panel validated the priority weights, caps, or utility. The study also does not measure extraction accuracy, device latency, memory, energy, provenance authenticity, or real-world harm. These limitations preclude policy or deployment claims. The RQ1 geographic anchors improve contextual plausibility, but the archived artifact does not contain the raw upstream snapshots, their source checksums, or per-row lineage. Consequently, Table 1 supports only a transparent account of construction, not independent provenance verification or representativeness of real rescue-report errors.

6 Conclusion

The study establishes conditional graph-localization bounds and exact-copy invariance of the family-wise priority score under fixed evidence grouping. Fixed clustering stresses reveal degradation under location and timestamp noise, and substantial degradation across all four graph variants at fivefold exact-copy multiplicity. Thus score-level invariance does not imply clustering invariance or end-to-end robustness. Together with the ranking and dispatch results, the evidence provides an auditable characterization of failure modes, while leaving general superiority, policy validity, and field readiness unestablished.

Data and Code Availability

The versioned artifact is archived on Zenodo [9] and corresponds to GitHub Release v1.0.1; the tag identifies the exact source snapshot and the concept DOI

the latest archive. It includes suite manifests, notebooks, locked configurations, per-run results, paired comparisons, the 40-seed RQ3 reanalysis, figure code, and result hashes. The exact RQ1 benchmark notebook/configuration snapshot is `6ac75c2`; the generator/external-benchmark package is retained at `a6be3e9`. The public `clean` branch at commit `5fad10d` additionally contains the primary fixed RQ1 stress artifact and an independent rerun under the same locked protocol, including per-run results, paired effects, and checksum manifests. The primary artifact includes its executed notebook; collection of the rerun’s executed notebook remains pending. These files are not part of release `v1.0.1`, and the two runs are not pooled as independent observations. Raw upstream anchor snapshots are not redistributed. Original code is MIT licensed. Publication and reuse of the contribution follow the applicable Springer publishing agreement; other non-code materials and third-party content retain their stated terms.

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